Software and Hardware Tools

Task 1 : Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Software Used: Wireshark

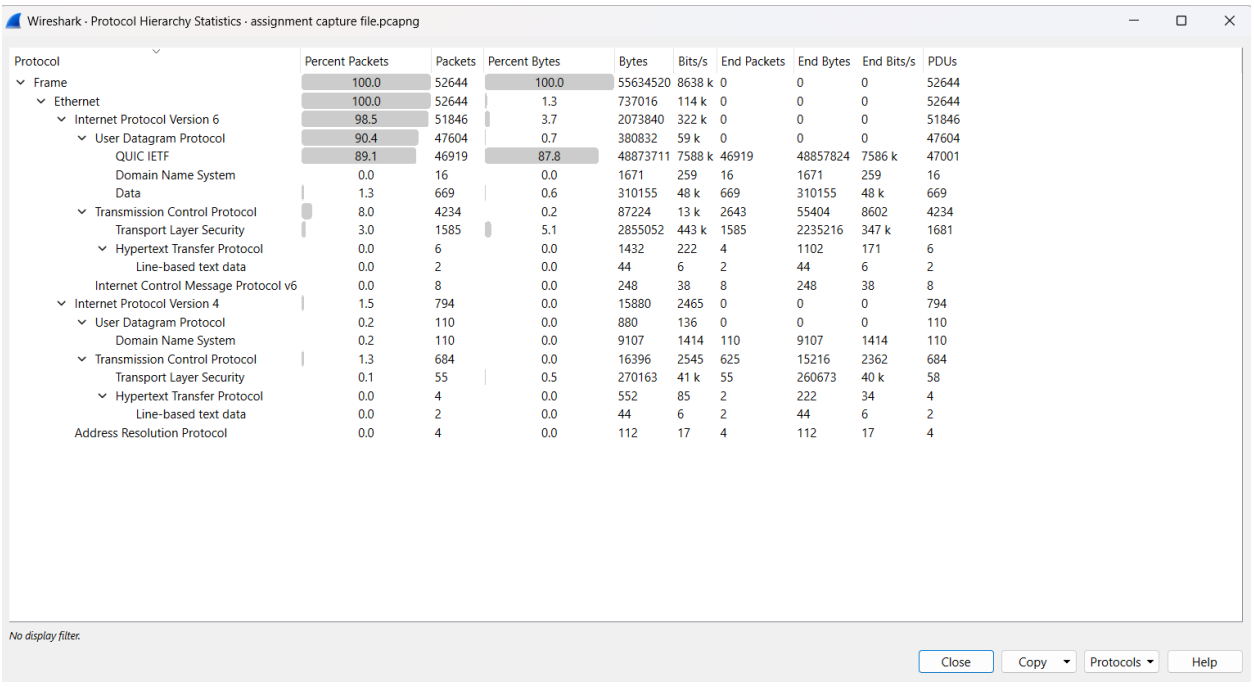
Procedure:

1. Installed Wireshark on the laptop.
2. Selected the active Wi-Fi interface.
3. Started capturing network traffic.
4. Browsed the internet while capturing packets.
5. Stopped the capture and saved the file.

Result:

- Total Packets Captured: 52,644
- Protocol 1: IPv6 – 51,846 packets
- Protocol 2: UDP – 47,604 packets
- Protocol 3: QUIC IETF – 46,919 packets
- Capture File: assignment capture file.pcapng

Screenshot:



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	52644	100.0	55634520	8638 k	0	0	0	52644
Ethernet	100.0	52644	1.3	737016	114 k	0	0	0	52644
Internet Protocol Version 6	98.5	51846	3.7	2073840	322 k	0	0	0	51846
User Datagram Protocol	90.4	47604	0.7	380832	59 k	0	0	0	47604
QUIC IETF	89.1	46919	87.8	48873711	7588 k	46919	48857824	7586 k	47001
Domain Name System	0.0	16	0.0	1671	259	16	1671	259	16
Data	1.3	669	0.6	310155	48 k	669	310155	48 k	669
Transmission Control Protocol	8.0	4234	0.2	87224	13 k	2643	55404	8602	4234
Transport Layer Security	3.0	1585	5.1	2855052	443 k	1585	2235216	347 k	1681
Hypertext Transfer Protocol	0.0	6	0.0	1432	222	4	1102	171	6
Line-based text data	0.0	2	0.0	44	6	2	44	6	2
Internet Control Message Protocol v6	0.0	8	0.0	248	38	8	248	38	8
Internet Protocol Version 4	1.5	794	0.0	15880	2465	0	0	0	794
User Datagram Protocol	0.2	110	0.0	880	136	0	0	0	110
Domain Name System	0.2	110	0.0	9107	1414	110	9107	1414	110
Transmission Control Protocol	1.3	684	0.0	16396	2545	625	15216	2362	684
Transport Layer Security	0.1	55	0.5	270163	41 k	55	260673	40 k	58
Hypertext Transfer Protocol	0.0	4	0.0	552	85	2	222	34	4
Line-based text data	0.0	2	0.0	44	6	2	44	6	2
Address Resolution Protocol	0.0	4	0.0	112	17	4	112	17	4

Task 2 : Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Commands Used:

ping www.spotify.com

ping www.zomato.com

Procedure:

1. Open Command Prompt.
2. Run the ping command for www.spotify.com.
3. Note down the average response time.
4. Repeat the same process for www.zomato.com.
5. Compare the average response times.

Result:

Website	Average Response Time
Spotify	58 ms
Zomato	48 ms

- Spotify Packet Loss: 0%
- Zomato Packet Loss: 0%
- Faster Website: Zomato

```
C:\Users\Admin>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42::810] with 32 bytes of data:
Reply from 2a04:4e42::810: time=50ms
Reply from 2a04:4e42::810: time=51ms
Reply from 2a04:4e42::810: time=63ms
Reply from 2a04:4e42::810: time=69ms

Ping statistics for 2a04:4e42::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 50ms, Maximum = 69ms, Average = 58ms

C:\Users\Admin>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:1e00::1737:f4c3] with 32 bytes of data:
Reply from 2600:140f:1e00::1737:f4c3: time=34ms
Reply from 2600:140f:1e00::1737:f4c3: time=48ms
Reply from 2600:140f:1e00::1737:f4c3: time=62ms
Reply from 2600:140f:1e00::1737:f4c3: time=51ms

Ping statistics for 2600:140f:1e00::1737:f4c3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 34ms, Maximum = 62ms, Average = 48ms
```

Task 3 : Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Tool Used: Windows Task Manager

Procedure:

1. Press Ctrl + Shift + Esc to open Task Manager.
2. Open the Processes tab.
3. Expand Google Chrome to view its individual processes.
4. Note down the process names and memory usage.

Result:

Google Chrome – 18 Processes

Process	**Memory Usage**
Chrome 1	2.2 MB
Chrome 2	2.2 MB
Chrome 3	6.4 MB
Chrome 4	0.6 MB
Chrome 5	404.9 MB
Chrome 6	2.5 MB
Chrome 7	276.5 MB
Chrome 8	21.8 MB
Chrome 9	111.2 MB
Chrome 10	411.6 MB
Chrome 11	56.7 MB
Chrome 12	29.2 MB
Chrome 13	24.0 MB
Chrome 14	7.3 MB

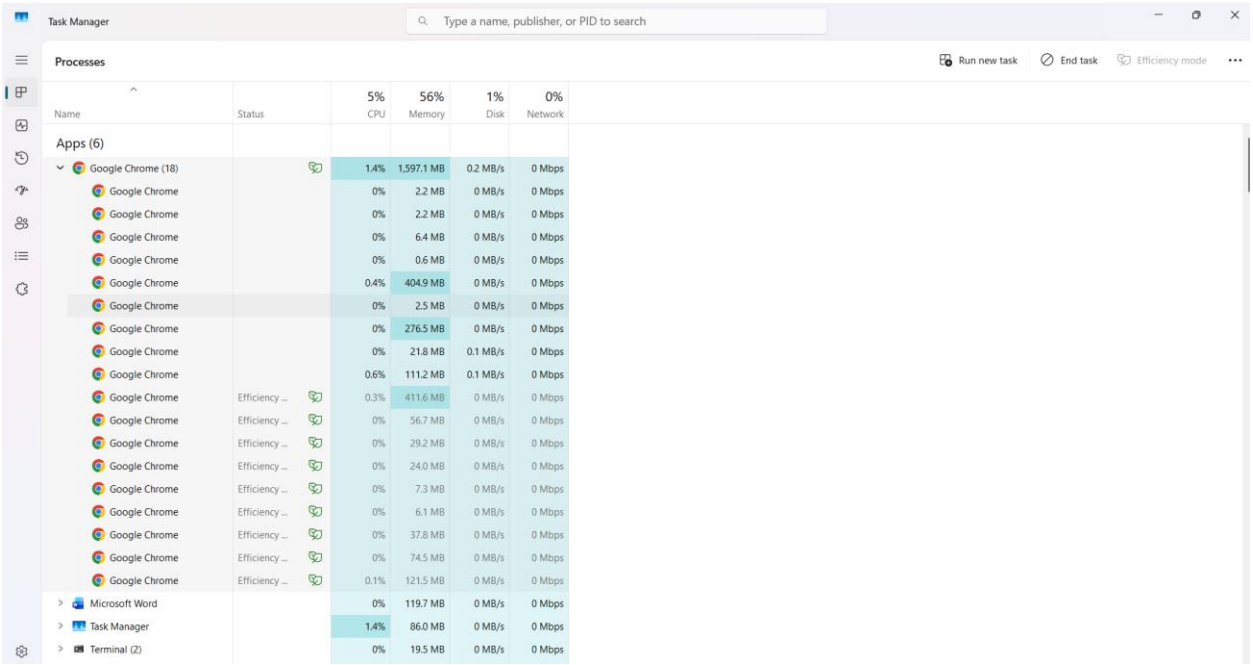
Chrome 15	6.1 MB
Chrome 16	37.8 MB
Chrome 17	74.5 MB
Chrome 18	121.5 MB

- Total Chrome Memory Usage: 1,597.1 MB
- WhatsApp Desktop: Not running
- Microsoft Edge: Not running

Conclusion:

Task Manager helps monitor running applications and their resource usage, including CPU, memory, disk, and network usage.

Screenshot:



Task 4 : Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Software Used: Angry IP Scanner

Procedure:

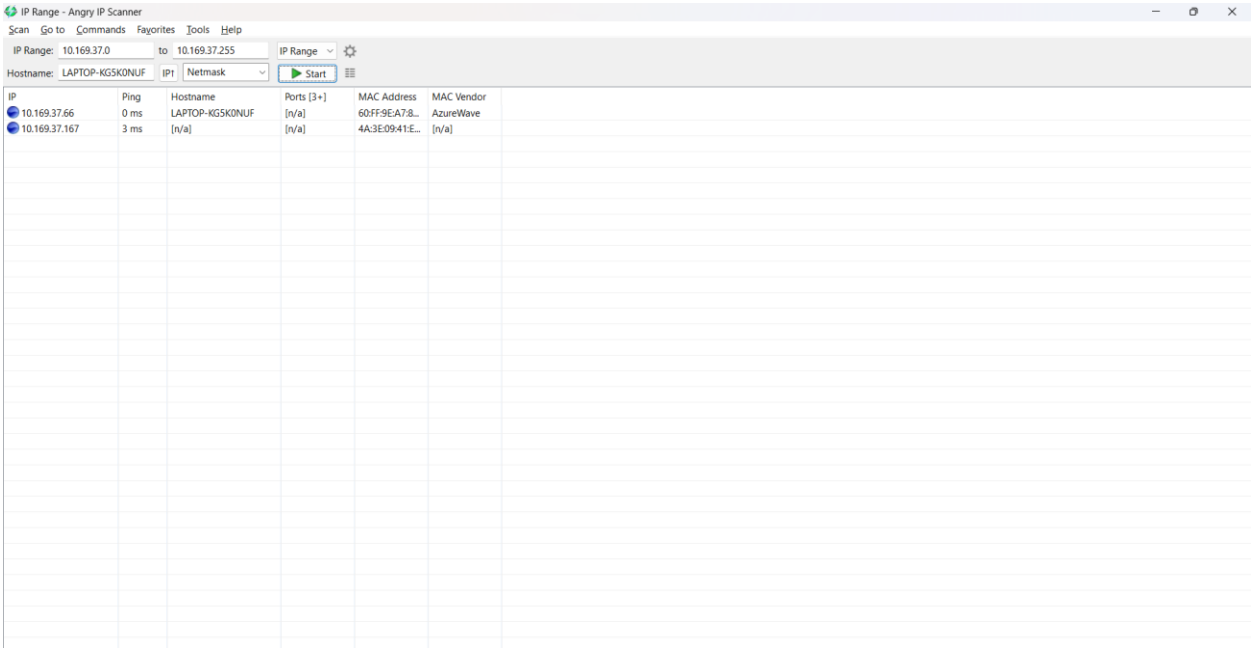
1. Downloaded and installed Angry IP Scanner.
2. Opened the application.
3. Entered the local network IP range (10.169.37.0 – 10.169.37.255).
4. Started scanning the network.
5. Observed the IP addresses and hostnames of active devices.

Result:

Device	IP Address	Device Name
Device 1	10.169.37.66	LAPTOP-KG5K0NUF
Device 2	10.169.37.167	Hostname Not Available

- Total Active Devices Found: 2

Screenshot:



Task 5 : Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Tools Suggested Using: ChatGPT

1. Cable Tester

A cable tester is used to check Ethernet cables for wiring faults, broken wires, and connectivity problems.

2. Crimping Tool

A crimping tool is used to attach RJ45 connectors to Ethernet cables when making or repairing network cables.

3. Digital Multimeter

A digital multimeter is used to measure voltage, resistance, and continuity when troubleshooting power-related hardware problems.

Conclusion:

These hardware tools help network technicians identify and fix physical network problems, test cables, and troubleshoot hardware connections.